CHE301 Midterm (VU Biology Zone)

CHE301 MIDTERM SOLVED

PAST PAPERS

1. **What do you know about pareto diagram?**

•Pareto Diagram is a combination of bar and line graphs of accumulated data. It is used where data associated with a problem.The name "Pareto" came from an Italian mathematician who created the diagram

2. **Name the sources of errors?**

Instrumental error, Method error, Human error.

3. **Goals of sampling?**

SampleSize, Sampling types, Samplingtechniques, Samplepreparations, To obtain a mean analyte concentration, Unbiased estimate of the population mean.

4. **What is limit of quantitation?**

It is the lowest concentration of an analyte in a sample that may be determined with acceptable accuracy and precision.

5. **How equipment can be calibrated?**

Calibration of an analytical balance using standard weights, Calibration of pipette / micropipettes and other volumetric apparatus to SI units using a calibrated balance, Calibration of pH meter with standard buffers.

6. **Give the sequence of quality assurance from SI units to products?**

Quality Assurance- Considerations Prior to Method Validation

Suitability of Instrument- Status of Qualification and Calibration

Suitability of Materials- Status of Reference Standards, Reagents, etc.

Suitability of Analyst- Status of Training and Qualification Records

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Create By Anaiza Anjum. Remember in your prayers.

Suitability of Documentation- Written analytical procedure and proper approved protocol with pre-established acceptance criteria.

7. Colloidal suspension crystalline suspension?

1-Colloidal suspensions • Particles are invisible to the naked eye (10-7 - 10-4 cm in diameter). • Colloidal particles show no tendency to settle from solution • not easily filtered.

2-Crystalline suspension • particles with dimensions on the order of tenths of a millimeter or greater. • The temporary dispersion of such particles of a tend to settle spontaneously,easily filtered.

8. What is error?

Error simply means not correct. But mathematically it may have some other meanings as well.

9. Types of sampling?

There are two major types of sampling Probability and Non-probability,which are further divided into sub types.

10.What factors needed for instrumental technique?

The instrumentshould be calibrated. Should follow properstandard operating procedure and standard testing methods. Instrumentshould be installed,operated and performedaccording to standard specifications. And the last but not least is statisticalevaluation of analytical data.

11.Molarity?

Molarity is the concentration of a particular chemical species. Formality, on the other hand, is a substance's total concentration without regard to its specific chemical form.

12.Method validation?

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13.Use of uv spectroscopy?

•UV Spectroscopy- To find wavelength, To estimate concentration of unknown in sample.

14.Use of Nmr Spectroscopy and mass spectrometry?

- NMR spectroscopy- On basis of magnetic properties define certain nuclei.
- Mass Spectrometry- Used to evaluate mass of separated molecules, Used as a detector.

15.what is beer lambert law?

Beer Lambert Law:

A=log10 Io/I = E/c

I0 = Incident light, I = Transmitted light, l = path length, C = Concentration, □ = Molar AbsorptivityConstant Allthe spectroscopic techniques use certainkindof electromagnetic radiationsand are based on Beer Lambert law.

16.Name two basic types of hypothesis?

Hypothesis type

- •Null Hypothesis- H0,
- •Alternate Hypothesis-Ha

17.what is standard deviation?

Standard deviation (the measure of precision), Reporting of data-Significant figures, Treating significant figures in calculations, Rounding off data, Confidence Interval, CV and SEM.

18.what is limit of quantitation?

Limit of Quantitation (LOQ): • It is the lowest concentration of an analyte in a sample that may be determined with acceptable accuracy and precision.

19.what is sample preparation?

Sample preparation Toanalyse we have to digest or extract our analyte

from the sample. This is called sample preparation.

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20.Explain the cause and effect diagram?

Cause & Effect Diagram

• Solving a problem requires clarification of a cause and effect relationship, where the effect (e.g., the result of work) varies according to factors (e.g., facilities, machines used, method of work, workers, materials and parts used etc.). • To obtain a good work result, we must identify the effects of various factors and develop measures to improve the result accordingly.

• Can obtain a clear overall picture of causal relation. (A change in the cause triggers a variation in the result.) • Can list up all causes to identify important causes. • Can determine the direction of action (counter measure).

21.Define median explain with example?

The middle of the reading in odd number of data is called median. If you have even number of data points then add the two mid readings and divide by two. Means take mean of the two middle reading.

22.which factor we consider in prior to method validation?

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23.what is standard deviation?

Standard deviation is measure of precision. It shows how close is your data in a data set.

It denote with sigma. Standard Deviation formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

24.what is limit of quantitation?

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25.what is limit of quantitation?

• Limit of Quantitation (LOQ):

• It is the lowest concentration of an analyte in a sample that may be determined with acceptable accuracy and precision.

26.write name of atomic emission spectroscopy?

Atomic Emission Spectroscopy

• ICP/OES and ICP MS

• Atomization source changed to inductive coupled plasma

• Temperature of plasma can reach up to 10,000K

• 70 elements can be determined to ppb level

• Low chemical interferences

27.what is assay of drug product?

Assay of Drug product

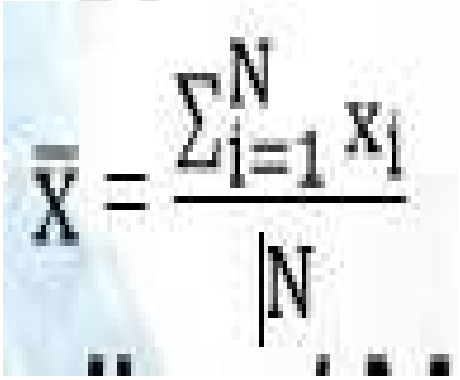
• Application of the analytical procedure to synthetic mixtures of the drug product components to which known quantities of the drug substance to be analyzed have been added.

28.What is mean?

Means with minimum variability in between of analytical results.

Mean

• arithmetic mean,s



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29.write the name of types of atomic Emission spectroscopy?

• Inductive Coupled Plasma –Optical Emission Spectroscopy (ICP / OES) • Inductive Coupled Plasma -Mass Spectrometry (ICP MS)

30.What is normality?

Normality defines concentration in terms of an equivalent, which is the amount of one chemical species reacting stoichiometrically with another chemical species.

31.write the main types of error?

Types of error

• Systematic error

• Random error

Remember us in your prayers.

Regards;

Admins & Moderators of **VU Biology Zone.**

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DRM Software Review